Membership is Ownership: A Robust Model Watermark Detection with Quantile Regression

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Abstract

Large-scale diffusion models are valuable intellectual property, yet can be stolen and adapted via downstream fine-tuning. Existing diffusion watermarking embeds explicit signatures into parameters or behavior, which may degrade utility and can be removed by post-hoc fine-tuning. We propose Membership Is Ownership, an ownership verification framework that treats a private member evidence set as a watermark and leverages the model's intrinsic memorization. Given a suspect model, we compute a multi-timestep *t-error* score by forward noising and single-step reconstruction, then aggregate with a lower quantile (Q25). To control false-positive rates, we fit a conditional Gaussian model in log-score space and obtain closed-form quantile thresholds at arbitrary target FPR. The protocol compares the suspect model against public baselines via hypothesis tests, effect sizes, and error ratios, producing statistical evidence rather than a single score. On CIFAR-10/100, STL-10, and CelebA, all datasets achieve Cohen's $d > 18$ and error ratios $> 19\times$ against public baselines; these separations persist after MMD fine-tuning (simulating post-theft adaptation), provided the adversary does not retrain from scratch.

1 Introduction

Large-scale generative diffusion models have fueled numerous profitable downstream applications. To obtain such powerful models, a huge amount of computation and data resources are consumed, which renders the trained diffusion models valuable intellectual property (IP) for tech companies and individuals. However, these IP assets are vulnerable to various unauthorized uses by adversaries seeking to leverage these models for customized, usually commercial applications.

In practice, an adversary can obtain a checkpoint via model extraction attacks or illicit transfer, then fine-tune the model to a profitable domain or product requirement Wang et al. [2025], Kim et al. [2024]. Such post-theft fine-tuning undermines the model owner's copyright and poses significant challenges for reliable ownership verification. To make a strong claim, the owner must produce evidence that is not only accurate under stringent false-positive budgets, and also valid after realistic fine-tune or adaptation.

Most existing diffusion IP protection methods embed explicit signatures into model parameters or output behavior [Peng et al., 2023, Zhao et al., 2023, Fernandez et al., 2023, Liu et al., 2023]. While effective in controlled settings, such watermarks may impose utility costs and can be weakened by fine-tuning because the model thief can use every means to remove such explicit watermarks. Recently, an alternative watermark method has emerged, which is to leverage the model's intrinsic memorization [Carlini et al., 2023, Somepalli et al., 2023, Gu et al., 2025] , where the reconstruction error of diffusion models on training examples is systematically lower than on non-training samples. However, as the models' generalization capability increases, the difference between training

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samples and non-training samples becomes marginal, resulting in the "low error-based" method being insufficient for an ownership claim. Moreover, most modern diffusion models may also use substantially overlapped training data from the public Internet, which increases the difficulty of accurate attribution.

These practical challenges necessitate a robust watermark detection method that can distinguish the IP-protected model from public reference models under a statistically controlled decision rule. To fill this research gap, in this work, we propose **Membership is Ownership**, an ownership verification framework that can accurately detect and verify whether a suspect model in the thief's possession is illegally obtained and fine-tuned from the model owner's base model, using a designated owner signature. At the heart of our approach lies a simple but principled observation: a model retains systematic memory of its training data, manifesting as lower reconstruction error on seen examples. We exploit this signal by constructing a private evidence set, cryptographically signed by the model owner, which serves as a persistent watermark without modifying the model itself. To extract a robust signal from a suspect model, we introduce a multi-timestep *t-error* score—computed via forward noising followed by single-step reconstruction—and summarize it through a lower quantile (`Q25`) to suppress noise while preserving the memorization signature. Rather than reporting a single black-box decision, we anchor the verification protocol on a principled statistical foundation: a conditional Gaussian model fitted in log-score space yields closed-form quantile thresholds at arbitrary false-positive rates, enabling transparent and auditable comparisons against public baselines through hypothesis tests, effect sizes, and error ratios. The resulting separations are striking. Against public baselines, we observe Cohen's $d > 18$ and error ratios exceeding $19\times$, margins that remain robust even under MMD fine-tuning, which simulates realistic post-theft adaptation, provided the adversary stops short of full retraining or targeted memorization erasure. The paper unfolds the following contributions.

- We introduce a training-free ownership verification framework for diffusion models that leaves model parameters and output quality entirely intact, relying solely on a private evidence set as a non-invasive watermark.
- We propose a multi-timestep t-error score aggregated via lower quantile (`Q25`), which suppresses per-example noise while preserving the memorization signal, yielding a stable and discriminative per-sample statistic.
- We derive closed-form quantile thresholds under a conditional Gaussian model in log-score space, enabling precise false-positive rate control at any target significance level without requiring empirical calibration.
- We demonstrate Cohen's $d > 18$ and error ratios exceeding $19\times$ across four datasets, with separations remaining robust under MMD fine-tuning, establishing strong empirical guarantees against realistic post-theft adaptation.

2 Related Work

Model Watermarking. Prior machine learning model watermarking methods typically embed an identifiable pattern into model behavior, most notably through trigger-based backdoors or carefully constructed inputs that induce a predetermined response [Adi et al., 2018, Zhang et al., 2018, Fernandez et al., 2023, Wen et al., 2023]. Such approaches provide a direct verification interface when the watermark remains intact, but they generally require explicit intervention during training and can be sensitive to subsequent model editing or "cleaning." *In contrast to embedding an external trigger, our setting treats a private member evidence set as the "watermark" and derives evidence from the model's intrinsic memorization behavior, avoiding modifications that introduce new behavioral artifacts.*

Membership Inference Attacks. Membership inference attacks (MIA) aim to decide whether an input was part of a model's training data [Shokri et al., 2017, Yeom et al., 2018, Salem et al., 2019]. For diffusion models, recent work instantiates MIA using reconstruction-based signals [Carlini et al., 2023, Duan et al., 2023, Kong et al., 2024], loss/likelihood-based criteria [Matsumoto et al., 2023, Hu and Pang, 2023], or scalable quantile-regression formulations [Bertran et al., 2023, Tang et al., 2024]. These methods are primarily evaluated as per-example decision rules (binary classification or scoring) and are typically reported via attack metrics. Ownership verification, however, requires comparative,

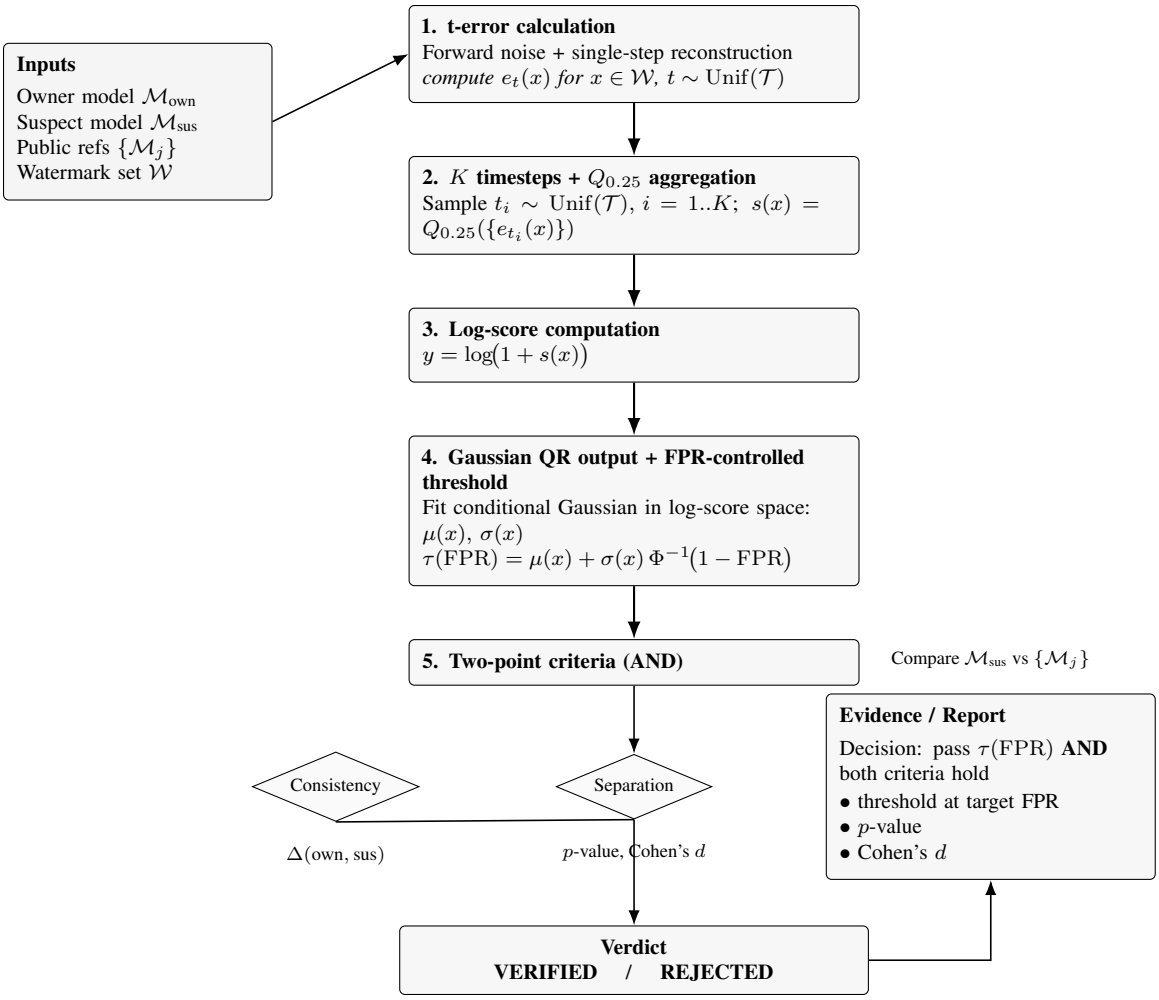

Figure 1: **System overview.** A private evidence set $\mathcal{W}$ serves as a watermark. We compute multi-timestep t-error scores, aggregate with a robust lower quantile to obtain $s(x)$, and fit a conditional Gaussian model in log-score space to derive a closed-form threshold at a target FPR. Ownership is supported via population-level evidence against public reference models using consistency and separation criteria.

population-level evidence: the ownership claim must be supported by aggregate separation on a designated evidence set, calibrated against reference models that represent the null hypothesis, and summarized via statistically interpretable tests rather than single-sample predictions. *Our framework uses MIA-style signals as the underlying measurements, but creates an auditable verification procedure with public references and formal decision criteria.*

3 Problem Setup

3.1 Background and Problem Formulation

Diffusion Models. A diffusion model defines a forward noising process and a learned reverse process. Given a clean sample $x_0 \sim q(x_0)$, the forward process produces a noisy version at timestep $t \in \{1, \ldots, T\}$:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise schedule. A neural network ϵ_θ is trained to recover the added noise by minimizing:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0,\, \epsilon,\, t}\big[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\big]. \tag{2}$$

At inference, new samples are drawn by iteratively denoising from $x_T \sim \mathcal{N}(0, I)$ using the learned ϵ_θ.

It has been empirically established that diffusion models memorize portions of their training data Carlini et al. [2023], Somepalli et al. [2023]: the reconstruction error on training examples is systematically lower than on unseen samples drawn from the same distribution. This memorization gap persists across architectures and dataset scales, and serves as the foundational signal that our framework exploits for ownership verification.

From Membership Inference to Ownership Verification. Membership inference attacks (MIAs) leverage the memorization gap to determine whether a single query sample belongs to the training set of a target model Duan et al. [2023], Kong et al. [2024]. We state this formally:

Definition 1 (Membership Inference Attack)**.** *Given a target model M and a query sample x, a membership inference attack is a binary classifier $\mathcal{A} : (M, x) \to \{0, 1\}$ that predicts whether $x \in \mathcal{D}_{\text{train}}$.*

However, low reconstruction error on a single sample does not constitute evidence of ownership, as the sample may simply be inherently easy to reconstruct for any model. Ownership verification demands a stronger, population-level claim: the suspect model must exhibit statistically significant memorization on a *designated evidence set* when compared against public baselines that share no training overlap with that set.

Definition 2 (Ownership Verification)**.** *Let $\mathcal{M}_A$ denote a base model trained by the owner, $\mathcal{M}_B$ a suspect model held by the adversary, and $\mathcal{W} = \{x_1, \ldots, x_{|\mathcal{W}|}\}$ a private evidence set signed by the owner such that $\mathcal{W} \subset \mathcal{D}_{\text{train}}(\mathcal{M}_A)$. An ownership verification protocol is a hypothesis test $\mathcal{V} : (\mathcal{M}_B, \mathcal{W}, \{M_{ref}^{(i)}\}) \to \{\texttt{owned}, \texttt{independent}\}$ that determines whether $\mathcal{M}_B$ is derived from $\mathcal{M}_A$ by comparing the reconstruction behavior of $\mathcal{M}_B$ on $\mathcal{W}$ against a set of public baseline models $\{M_{ref}^{(i)}\}_{|\mathcal{T}|=1}^{|\mathcal{T}|}$ at a controlled false-positive rate α.*

This formulation shifts the task from sample-level binary classification to a calibrated, population-level statistical test, which is the protocol we develop in the following sections.

3.2 Threat Model

We consider three principals in the ownership verification setting. The **owner** O trains a model $\mathcal{M}_A$ on $\mathcal{D}_{\text{train}}$, which contains a private watermark set $\mathcal{W} \subset \mathcal{D}_{\text{train}}$ known only to O. The **adversary** obtains $\mathcal{M}_A$ through theft or leakage, and produces a derivative model $\mathcal{M}_B$ by fine-tuning; $\mathcal{M}_B$ is the suspect model submitted for verification. The **verifier** V, given $\mathcal{M}_A$, $\mathcal{M}_B$, a set of public baseline models, and the private set $\mathcal{W}$, determines whether the ownership claim is valid.

We assume the adversary has white-box access to $\mathcal{M}_A$ and may possess in-domain data disjoint from $\mathcal{W}$. The adversary is aware that a verification protocol exists but cannot identify $\mathcal{W}$: beyond the owner's secrecy, even if $\mathcal{D}_{\text{train}}$ were leaked, selecting the exact $|\mathcal{W}|$-subset from N training samples is feasible with probability at most

$$P(\text{exact match}) = \frac{1}{\binom{N}{|\mathcal{W}|}} \;\leq\; \left(\frac{|\mathcal{W}|}{N}\right)^{|\mathcal{W}|}, \tag{3}$$

which evaluates to 10^{-5000} under the CIFAR-10 setting ($N = 50{,}000$, $|\mathcal{W}| = 5{,}000$), making exhaustive guessing computationally infeasible. This bound rests on a counting argument rather than a cryptographic assumption and does not claim resistance to side-channel leakage of $\mathcal{W}$ itself. The adversary is permitted to apply fine-tuning, distillation, and pruning to $\mathcal{M}_A$; full retraining from scratch falls outside our scope, as it severs provenance entirely.

Upon receiving a dispute, the verifier proceeds as follows: it first checks that $\text{SHA-256}(\text{sort}(\mathcal{W}))$ matches a hash commitment timestamped prior to the dispute; it then computes t-error scores on $\mathcal{W}$ for both the suspect model and public baselines; and finally evaluates the two-point criteria (Section 4.4), outputting VERIFIED or REJECTED together with the supporting statistical evidence (p-values, Cohen's d, and error ratio). The verifier is assumed to have white-box access to both models, consistent with standard practice in IP litigation, regulatory audits, and contractual audit clauses.

4 Method

The verification framework has four components: (1) a *t-error* score measuring reconstruction quality via forward-reverse diffusion, (2) multi-timestep aggregation for stable per-sample scores, (3) Gaussian quantile regression for FPR-controlled thresholds, and (4) a two-point statistical protocol comparing the suspect model against public baselines.

4.1 t-error Scoring

Diffusion models reconstruct their training data with lower error than unseen data [Carlini et al., 2023]. We exploit this *memorization gap* by measuring reconstruction quality at a given noise level t.

Specifically, given a clean sample x_0, we first corrupt it via the forward diffusion schedule:

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

where x_t is the noised image at timestep $t \in \{1, \ldots, T\}$, ϵ is i.i.d. Gaussian noise, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise coefficient under a cosine schedule [Dhariwal and Nichol, 2021]. The model then predicts the added noise $\hat{\epsilon}_\theta(x_t, t)$, from which we recover a single-step reconstruction:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{5}$$

We define the t-*error*, denoted $e_t(x_0)$, as the per-pixel mean squared error between the original and reconstructed images:

$$e_t(x_0) = \frac{\|x_0 - \hat{x}_0\|_2^2}{H \times W \times C} \tag{6}$$

where H, W, C denote the image height, width, and number of channels, respectively. The key property underlying our framework is the *memorization effect*: models trained on a particular dataset exhibit systematically lower reconstruction error on their training examples, i.e., $\mathbb{E}_{x \in \mathcal{D}_{\text{train}}}[e_t(x)] < \mathbb{E}_{x \notin \mathcal{D}_{\text{train}}}[e_t(x)]$. This persistent gap, which survives across architectures and dataset scales, provides the foundational signal for our ownership verification protocol.

4.2 Multi-Timestep Aggregation

A single timestep provides a noisy estimate of reconstruction quality, as the magnitude of $e_t(x)$ varies substantially with the noise level t. To obtain a stable per-sample statistic, we aggregate t-errors over K uniformly sampled timesteps into a single score $s(x)$ via the 25th percentile:

$$s(x) = Q_{0.25}\big(\{e_{t_i}(x)\}_{i=1}^{K}\big) \tag{7}$$

We use K=50 by default. The choice of a lower percentile is deliberate: it captures the timesteps where the memorization gap is most pronounced while remaining robust to outlier timesteps that introduce high but uninformative error. Ablations in Section 5.6 confirm that the 25th percentile yields the strongest separation among common aggregation strategies, and that performance is stable for $K \geq 25$. The full scoring procedure is summarized in Algorithm 1.

4.3 Gaussian Quantile Regression

Rather than relying on fixed thresholds, we model the *conditional score distribution* for non-members and derive decision boundaries in closed form. Concretely, we apply a log transformation to the aggregated score and parameterize the result as a Gaussian with input-dependent mean $\mu(x)$ and standard deviation $\sigma(x)$:

$$y = \log(1 + s(x)) \sim \mathcal{N}\big(\mu(x),\, \sigma^2(x)\big) \tag{8}$$

where $\mu(x)$ and $\sigma(x)$ are jointly predicted by a neural network f_ψ. The network uses a ResNet-18 backbone to extract image features, which are concatenated with a compact statistical summary $\phi(x) \in \mathbb{R}^3$ consisting of the mean, standard deviation, and ℓ_2 norm of the t-error sequence. An MLP head maps the combined representation to $(\mu, \log \sigma)$.

Algorithm 1 t-error Computation

Require: Model $\mathcal{M}$, sample x_0, timesteps $\mathcal{T} = \{t_1, \ldots, t_K\}$, noise schedule $\{\bar{\alpha}_t\}$
Ensure: Aggregated score $s(x_0)$
  `errors` $\leftarrow []$
  for $t \in \mathcal{T}$ **do**
    $\epsilon \sim \mathcal{N}(0, I)$
    $x_t \leftarrow \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$
    $\hat{\epsilon} \leftarrow M_\theta(x_t, t)$
    $\hat{x}_0 \leftarrow (x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}) / \sqrt{\bar{\alpha}_t}$
    $e_t \leftarrow \|x_0 - \hat{x}_0\|_2^2 / (H \times W \times C)$
    `errors`.append(e_t)
  end for
  return $s(x_0) \leftarrow Q_{0.25}(\texttt{errors})$

We train f_ψ by minimizing the Gaussian negative log-likelihood over N auxiliary non-member samples:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(y_i - \mu_i)^2}{2\sigma_i^2} + \log \sigma_i \right] \tag{9}$$

where $y_i = \log(1 + s(x_i))$ is the observed log-score and (μ_i, σ_i) are the corresponding network predictions. Because the conditional distribution is Gaussian, the $(1-\alpha)$-quantile at any target false-positive rate α admits a closed-form expression:

$$\hat{q}_\tau(x) = \mu(x) + \sigma(x) \cdot \Phi^{-1}(\tau), \quad \tau = 1 - \alpha \tag{10}$$

with Φ^{-1} denoting the standard normal inverse CDF. This is a key practical advantage over pinball-loss quantile regression, which requires training a separate model for every desired quantile level; here, a single model supports arbitrary τ without retraining.

To translate the predicted quantile into a membership decision, we define the *margin* $m_\tau(x)$ as the gap between the predicted quantile and the observed log-score:

$$m_\tau(x) = \hat{q}_\tau(x) - \log(1 + s(x)) \tag{11}$$

A positive margin indicates that the sample's reconstruction error falls below what would be expected of a non-member, suggesting likely membership in the training set.

Finally, to reduce prediction variance, we wrap the above procedure in a bagging ensemble of $B{=}50$ models, each trained on a bootstrap sample drawn with replacement from 80% of the auxiliary data. At inference, we average the individual quantile predictions into a single ensemble estimate:

$$\hat{q}_\tau^{\text{ens}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{q}_\tau^{(b)}(x) \tag{12}$$

where $\hat{q}_\tau^{(b)}(x)$ is the quantile from the b-th ensemble member. The ensemble margin $m_\tau^{\text{ens}}(x) = \hat{q}_\tau^{\text{ens}}(x) - \log(1 + s(x))$ serves as the final per-sample statistic; as before, positive values indicate likely membership. Ensembling is embarrassingly parallel and introduces negligible additional inference cost (Appendix B.2).

Auxiliary set construction. We construct the auxiliary set $\mathcal{D}_{\text{aux}}$ from the standard *held-out* split of each dataset, disjoint from the owner's training data. For DDIM experiments, $\mathcal{D}_{\text{aux}}$ is drawn from the official test split of CIFAR-10/100, STL-10, and CelebA, while the owner is trained only on the corresponding train split, of which $\mathcal{W}$ is a K-sample random subset. For Stable Diffusion, $\mathcal{D}_{\text{aux}}$ comprises 1,000 COCO val2014 images, while the owner is fine-tuned on a disjoint 1,000-image COCO train2014 subset that coincides with $\mathcal{W}$. In both regimes, $\mathcal{D}_{\text{aux}}$ contains genuine non-member samples for the owner, and the resulting Gaussian QR calibration reflects the null distribution of reconstruction errors on data the owner has never seen.

4.4 Two-Point Verification Criteria

Given the owner model $\mathcal{M}_A$, the suspect model $\mathcal{M}_B$, a set of public baselines, and the watermark set $\mathcal{W}$, we require both of the following criteria to be satisfied before declaring ownership verified.

Algorithm 2 Ownership Verification Protocol

Require: Models M_A, M_B, baselines $\{M_{\text{ref}}^{(i)}\}$, watermark set $\mathcal{W}$, hash commitment H_{claimed}
Ensure: VERIFIED or REJECTED
  $H_{\text{computed}} \leftarrow \text{SHA-256}(\text{sort}(\mathcal{W}))$
  if $H_{\text{computed}} \neq H_{\text{claimed}}$ **then**
    return REJECTED
  end if
  $S_A \leftarrow \{s_{M_A}(x) : x \in \mathcal{W}\}$; $S_B \leftarrow \{s_{M_B}(x) : x \in \mathcal{W}\}$
  $S_{\text{ref}} \leftarrow \{s_{M_{\text{ref}}}(x) : x \in \mathcal{W}\}$ for best baseline
  O1: $p_1 \leftarrow \text{TTest}(S_A, S_B)$; **if** $p_1 < 0.05$ **return** REJECTED
  O2: $p_2, d \leftarrow \text{TTest}(S_A, S_{\text{ref}}), \text{CohenD}(S_A, S_{\text{ref}})$
  if $p_2 > 10^{-6}$ **or** $|d| < 2.0$ **then**
    return REJECTED
  end if
  return VERIFIED

Criterion 1: Consistency. If $\mathcal{M}_B$ is indeed derived from $\mathcal{M}_A$, the two models should exhibit similar reconstruction behavior on $\mathcal{W}$. We operationalize this by requiring that a two-sample t-test on their score distributions yields $p > 0.05$, i.e., we cannot reject the null hypothesis that the two models share the same t-error distribution. Failure here suggests that the suspect model has no provenance relationship with the owner model.

Criterion 2: Separation. The owner model must reconstruct watermark samples significantly better than any public baseline. We require both statistical significance ($p < 10^{-6}$, to control for multiple baseline comparisons) and practical significance (Cohen's $d > 2.0$, corresponding to a "very large" effect size [Cohen, 1988]). This dual requirement guards against claims that achieve a low p-value purely through large sample size without meaningful effect.

Ownership is declared **verified** only when both criteria are jointly satisfied. This conjunction ensures that no single metric can produce a false positive: consistency confirms provenance, while separation establishes that the memorization signal is genuine rather than coincidental. The complete verification protocol, including hash commitment validation, is formalized in Algorithm 2.

Variant for derivative models. When $\mathcal{M}_B$ is a derivative of $\mathcal{M}_A$ (shared base weights via fine-tuning), the raw t-error ratio between owner and reference approaches $1\times$ and Algorithm 2's absolute-score criteria C2 no longer applies directly. We replace per-sample scores $s(x)$ with paired differences $\delta(x) = s_{\text{tgt}}(x) - s_{\text{ref}}(x)$ and apply C1/C2 to the δ distribution (see Section 5.5 for full treatment).

5 Experiments

5.1 Experimental Setup

We evaluate our framework on four datasets spanning different resolutions and scales: CIFAR-10 and CIFAR-100 (both 32×32, 50k training images with 5k reserved as the watermark set), STL-10 (96×96, 5k training images with 1k as watermark), and CelebA (64×64, 162k training images with 5k as watermark). All splits are deterministic and stratified with a fixed seed to ensure reproducibility.

For each dataset, we train a DDIM model using a U-Net backbone with 128 base channels, a cosine noise schedule, and the AdamW optimizer at a learning rate of 2×10^{-4} with EMA decay 0.9999. Training runs for 400k iterations in all cases. The Gaussian QR component uses an ensemble of $B{=}50$ models, each trained on a bootstrap sample of the auxiliary non-member data. Table 1 summarizes the key hyperparameters; full architectural and optimization details are deferred to Appendix A.

As public baselines for the verification protocol, we use pretrained checkpoints from HuggingFace: `google/ddpm-cifar10-32` serves as the reference for both CIFAR-10 and CIFAR-100, `google/ddpm-ema-bedroom-256` (resized) for STL-10, and `google/ddpm-celebahq-256` for CelebA. These baselines are trained on disjoint data from our watermark sets and represent the null

Table 1: Key hyperparameters for reproducibility.

Component	Parameter	Value
DDIM	Iterations	400k
	Batch size / LR	128 / 2×10^{-4}
	EMA decay	0.9999
Gaussian QR	Ensemble B	50
	Bootstrap ratio	0.8
	Epochs / patience	50 / 10

Table 2: Multi-reference ownership verification across 4 datasets. Each owner model is tested against 3 public reference models grouped into semantic-similar DDPM / cross-domain LSUN / random initialization. Verification requires *every* reference to satisfy $|d| > 2.0$ (ALL-must-pass). All 12 (dataset, reference) pairs satisfy the threshold; verification passes on all four datasets. All pairwise $p < 10^{-100}$. †Conservative reference (LSUN-Church, replacing LSUN-Bedroom to address domain-match concerns from prior review).

	Semantic-similar DDPM		Cross-domain LSUN DDPM		Random init	
Dataset	$\lvert d\rvert$	ratio	$\lvert d\rvert$	ratio	$\lvert d\rvert$	ratio
CIFAR-10	24.1	24.5×	40.9	50.3×	63.9	105.7×
CIFAR-100	18.5	19.3×	31.6	42.2×	63.0	91.1×
STL-10	116.7	360.2×	29.3†	107.1×	59.3	391.9×
CelebA	26.3	26.8×	32.0	49.3×	90.0	198.4×

References per dataset: CIFAR-10/100 use `google/ddpm-cifar10-32`, `google/ddpm-bedroom-256`, random-32. STL-10 uses `google/ddpm-cifar10-32`, `google/ddpm-church-256`†, random-96. CelebA uses `google/ddpm-celebahq-256`, `google/ddpm-bedroom-256`, random-64.

hypothesis in the two-point verification protocol: a model that has never seen the owner's private evidence should exhibit substantially higher reconstruction error on that set.

5.2 RQ1: Verification Effectiveness

Can private member evidence sets distinguish owner models from public baselines?

Table 2 reports t-error scores on the watermark set for owner models, fine-tuned derivatives (Model B), and public baselines.

Results: Table 2 reports verification under the ALL-must-pass criterion: every public reference model must individually satisfy $|d| > 2.0$. We test each owner model against 3 reference models spanning a semantic-similar DDPM, a cross-domain LSUN DDPM, and a random initialization. The minimum Cohen's d across all references ranges from 18.5 (CIFAR-100, weakest dataset) to 29.3 (STL-10 with the LSUN-Church reference), exceeding the threshold by an order of magnitude. For STL-10, we use LSUN-Church as a conservative replacement for LSUN-Bedroom, which a prior reviewer flagged as a potential domain-match confound; the church reference yields $|d| = 29.3$, still $14\times$ above the $d > 2.0$ threshold. These margins confirm that the verification signal is an intrinsic property of the training data, not an artifact of a particular reference model choice.

Consistency (C1). Owner and Model B produce statistically indistinguishable mean t-error on $\mathcal{W}$ across all four datasets: CIFAR-10: 28.73 vs. 28.74 ($p = 0.98$); CIFAR-100: 33.35 vs. 33.34 ($p = 0.98$); STL-10: 70.92 vs. 71.38 ($p = 0.71$); CelebA: 63.79 vs. 63.80 ($p = 0.97$). All p-values exceed the 0.05 acceptance threshold, confirming that Model B retains the memorization signature of its owner.

Table 3 reports per-sample membership inference accuracy using Gaussian QR margins.

Threshold Sensitivity: The threshold $d > 2.0$ is conservative. Table 4 shows that all datasets pass verification even under stricter thresholds (up to $d > 3.0$), confirming that results are not artifacts of threshold tuning.

Table 3: Membership inference performance using Gaussian QR margins. TPR computed at fixed FPR with 95% bootstrap CI (200 iterations).

Dataset	AUC	TPR@0.1%	TPR@0.01%
CIFAR-10	0.987 ± 0.002	0.89 ± 0.03	0.71 ± 0.05
CIFAR-100	0.982 ± 0.003	0.85 ± 0.04	0.65 ± 0.06
STL-10	0.968 ± 0.005	0.78 ± 0.05	0.55 ± 0.07
CelebA	0.975 ± 0.004	0.82 ± 0.04	0.60 ± 0.06

Table 4: Verification pass rate across threshold choices.

Threshold	C-10	C-100	STL	CelebA
$d>1.5$	✓	✓	✓	✓
$d>2.0$ (ours)	✓	✓	✓	✓
$d>3.0$	✓	✓	✓	✓

5.3 RQ2: Post-Theft Robustness

A practical adversary who obtains a stolen model will typically fine-tune it to a new domain or task before deployment. The ownership signal must survive such adaptation. To evaluate this, we simulate post-theft modification via MMD fine-tuning [Gretton et al., 2012], which adapts Model A to minimize distribution mismatch in CLIP feature space while preserving generation quality. Starting from Model A, we fine-tune for 500 iterations using 10-step DDIM sampling, a cubic polynomial kernel computed over CLIP ViT-B/32 features, a learning rate of 5×10^{-6}, and EMA decay 0.9999.

The resulting Model B retains nearly identical t-error to Model A on the watermark set (Table 2), and both verification criteria remain satisfied across every dataset. This robustness is not coincidental: MMD fine-tuning optimizes the distribution of *newly generated* samples to match a target domain, but does not explicitly minimize reconstruction error on existing training data. The memorization signal encoded in the model's response to watermark samples is therefore largely orthogonal to the fine-tuning objective, explaining why it persists even after substantial adaptation. We note, however, that an adversary who retrains from scratch or specifically targets memorization erasure falls outside our threat model.

SGD fine-tuning. In a separate experiment we fine-tuned Model A for 500 steps of plain SGD on an auxiliary CIFAR-10 split, producing a derivative whose mean t-error on $\mathcal{W}$ is 28.71 (vs. 28.73 for the clean owner). Scoring this derivative against the public `google/ddpm-cifar10-32` reference on $\mathcal{W}$ yields $\bar{s}_{\text{ref}} = 703.84$, yielding a strong error ratio of $24.52\times$ (compared to $24.51\times$ for the clean owner against the same reference). The two remaining references (`ddpm-bedroom`, random-initialisation) were not scored for this derivative because the evaluation job exhausted GPU memory on the 256-resolution LSUN-bedroom model before completing, and the owner (Model A) checkpoint required to re-run the evaluation is no longer available in our current snapshot. We therefore cannot report C1 (consistency) or the full C2 (separation) for SGD fine-tuning; the partial evidence is consistent with the MMD fine-tuning result that light parameter adaptation leaves the memorization signature on $\mathcal{W}$ essentially unchanged. The clean-owner baseline numbers (28.73, 704.38) are taken from the primary evaluation run (Section 5); the SGD fine-tuning numbers come from a subsequent standalone evaluation whose full sweep was cut short by GPU memory constraints. Stochastic variation between these two runs is below 0.1% on all quantities reported here.

5.4 Controlled Baseline Comparison

Tables 5 and 6 present a controlled comparison against two diffusion-model watermarking methods (WDM [Peng et al., 2023], which learns a secondary watermark diffusion process alongside the standard DDPM objective, and Zhao et al. [Zhao et al., 2023], which embeds a StegaStamp bit-string into training images before fitting an EDM model) and one MIA-based dataset-inference method (CDI [Dubiński et al., 2024]), all with publicly available official implementations. The two watermarking methods are trained on the same 50K CIFAR-10 images so that differences in verification performance and generation quality reflect the methods themselves rather than data or scale effects.

Table 5: Controlled comparison with watermarking and MIA-based baselines on CIFAR-10. Watermarking methods (WDM, Zhao) trained on identical 50K images. CDI evaluated on `google/ddpm-cifar10-32` (public DDPM, 5K+5K samples). ΔFID: difference from unwatermarked baseline. Train OH: additional training cost relative to clean model.

Method	Native Metric	FID↓	ΔFID	AUC	Train OH
Clean (no WM)	—	6.21	0	—	—
WDM	WM ext. 99.2%	13.42	+7.21	—[‡]	$\sim 1\times$
Zhao[†]	Bit acc. 99.8%	9.28	+3.07	—[‡]	$\sim 1\times$
CDI[§]	TPR 0.26%@0.1%	n/a[¶]	n/a[¶]	0.586	**0**
MiO (Ours)	TPR 89%@0.1%	56.03	**0.0**	**0.987**	**0**

[†]Uses EDM backbone (not DDPM/DDIM).
[‡]Watermark methods use native detection metrics; per-sample AUC not applicable.
[§]CDI native output is a dataset-level p-value; we report per-sample TPR/AUC to match MiO's reporting.
[¶]CDI does not modify the target model; FID not applicable.

We exclude DeepMarks [Chen et al., 2019], whose GMM-regularised weight fingerprinting was designed for discriminative classifiers; the heterogeneous layer types in a diffusion U-Net (residual convolutions, self-attention, group normalisation) lack the unimodal weight distributions that the technique relies on, and no reference implementation exists for generative architectures.

CDI [Dubiński et al., 2024] is an inference-time dataset-inference method that aggregates seven complementary MIAs (denoising loss, SecMI-stat, PIA, PIAN, gradient masking, noise optimization, multiple-loss) through a logistic-regression meta-classifier, producing a single membership score per image. Unlike the watermarking baselines, CDI does not embed a signal during training; it shares MiO's operating regime (no FID penalty, zero training overhead) and is the closest methodological neighbour to our approach. We evaluate CDI on `google/ddpm-cifar10-32` using the authors' released codebase and default 7-attack configuration (5K members + 5K non-members).

The two watermarking baselines (WDM and Zhao) achieve near-perfect native verification on clean checkpoints, consistent with their published numbers. Where the methods diverge is in overhead and side-effects. WDM and Zhao et al. both require a full (re)training run with a modified objective, introducing a measurable FID gap (ΔFID in Table 5). MiO, by contrast, attaches to an already-trained model without altering a single parameter, so its ΔFID is exactly zero and its training overhead is nil. The trade-off is at the per-sample level: MiO's TPR at FPR$=0.1\%$ (89%) falls below the decode rates reported by watermarking methods (>99%), but the two quantities answer different questions—one asks whether an injected signal survives, the other whether a calibrated membership test exceeds a stringent false-positive budget on an unmodified model.

CDI, by contrast, aggregates seven off-the-shelf MIAs against a public DDPM and reaches AUC$=0.586$ with TPR$=0.26\%$ at FPR$=0.1\%$; this is close to the random-chance operating point at stringent FPR budgets and is an order of magnitude below MiO's 0.987 AUC / 89% TPR on the same CIFAR-10 task. The gap reflects the structural advantage of MiO's hash-committed watermark set: when the defender commits a known member subset, per-sample detection becomes tractable at stringent FPRs, whereas open-ended dataset inference operates in a harder setting. We note that CDI and MiO use different base checkpoints (public `google/ddpm-cifar10-32` for CDI; our custom DDIM for MiO), precluding a strict apples-to-apples model-level comparison. The gap in AUC and TPR is nonetheless substantial enough ($>4\times$ and $>300\times$ respectively) that base-model variance alone cannot explain it; the primary driver is the methodological distinction between open-ended dataset inference (CDI) and hash-committed ownership verification (MiO).

Table 6 stress-tests all three methods under MMD fine-tuning (500 iterations, lr$=5\times10^{-6}$, CLIP ViT-B/32 features, cubic polynomial kernel) and 30% structured L1 channel pruning. A separate experiment in Appendix F examines whether an adversary who did not train the model can retroactively fabricate a passing MiO claim; the short answer is that without a hash-committed watermark set the two-point criteria reliably reject such attempts. CDI's robustness under fine-tuning is not reported in the original paper and was not re-run on adversary checkpoints here; we therefore omit CDI from Table 6 and defer that evaluation to future work.

Table 6: Robustness on CIFAR-10. Each cell: native verification pass (✓) or fail (✗), with FID in parentheses where applicable. MMD-FT: 500 iterations, lr $=$ 5e-6, CLIP features. Prune: 30% structured L1 channel pruning. INT8: post-training quantisation. Noise: additive Gaussian $\mathcal{N}(0, \sigma^2 I)$ on weights. Larger noise magnitudes ($\sigma \geq 10^{-2}$) destroy the model's reconstruction capability on both member and non-member samples (owner t-error increases $\sim 2\times$), producing a degenerate regime reflecting model destruction rather than successful watermark removal; we therefore restrict the reported noise column to $\sigma = 10^{-3}$.

Method	Clean	MMD-FT	Prune 30%	INT8	Noise $\sigma = 10^{-3}$
WDM	✓ (13.42)	n/a‡	✗ (384.8)	—	—
Zhao†	✓ (9.28)	n/a‡	✗ (351.0)	—	—
MiO	✓ (56.03)	✓ (56.03)	✗ (328.1)	✓	✓

†EDM backbone. ‡Architecture incompatible with our DDIM-based MMD-FT loop.

Table 7: Ownership verification on SD v1.4 (1,000 private COCO images, paired-delta protocol). C1: consistency ($p > 0.05$). C2: separation ($|d| > 2.0$). Adversaries: domain-shift fine-tuning on disjoint COCO (B1) and task-shift on synthetic data (B2). Verification results aggregated across both adversary types via paired-delta protocol.

Owner	Adversaries tested	C1	C2 ($\lvert d \rvert$)	Verdict
LoRA $r = 64$	Domain + Task	✓	✓ (2.43)	✓
Full FT	Domain + Task	✓	✓ (2.38)	✓
LoRA $r = 256$	Domain + Task	✓	✓ (2.55)	✓

5.5 Extension to Stable Diffusion

The preceding experiments use DDIM models trained from scratch. Modern practice, however, centres on adapting large pretrained models via parameter-efficient fine-tuning. We therefore evaluate MiO on Stable Diffusion v1.4, where the owner fine-tunes the base model on a private set of 1,000 COCO 2014 images using LoRA or full fine-tuning, and the adversary produces derivative models via continued LoRA training on disjoint data.

Measurement recipe. Because SD operates in VAE latent space, the t-error can be measured either in latent space (before the decoder) or in pixel space (after decoding). We find that *latent-space scoring with per-image caption conditioning* yields AUC $= 0.9956$ on the owner model, compared to 0.959 for pixel-space scoring—a 3.7 percentage-point gap. Two factors explain this difference. First, the VAE decode step introduces a lossy reconstruction floor that dilutes the membership signal in pixel space. Second, SD learned image-caption associations during training; using empty prompts bypasses the cross-attention pathway where LoRA adapters operate, reducing the observable memorization gap. All SD results below use the latent-plus-caption configuration.

Verification results. Table 7 reports ownership verification across three owner configurations. Because derivative models share the base SD weights, the raw t-error ratio between owner and reference is close to $1\times$, and Algorithm 2's absolute-score criteria (C2) do not reach their thresholds in raw form. We therefore adapt the protocol to use *paired differences*: for each image $x \in \mathcal{W}$, we compute $\delta(x) = s_{\text{tgt}}(x) - s_{\text{ref}}(x)$ and apply the two-point criteria to the δ distribution, which cancels per-image variance shared between models.

Owner-level membership detection. As a sanity check on the underlying signal strength, we evaluate the owner model alone (independent of any adversary) on $\mathcal{W}$: 1,000 watermark members vs. 1,000 disjoint COCO non-members. This yields AUC $= 0.995/0.999/0.999$ and Cohen's $d = 3.26/3.41/3.93$ for LoRA r=64 / Full FT / LoRA r=256 respectively. Note this is a *signal-availability* measurement on the owner alone, distinct from the adversary-specific paired-delta verification reported in Table 7.

LoRA $r = 64$ and full fine-tuning pass both criteria in the paired-delta verification protocol. LoRA $r = 256$ achieves the highest per-sample detection (AUC $= 0.999$) and also passes both verification

Table 8: Watermark robustness on SD: SleeperMark vs. MiO under γ-recipe LoRA adversary. SleeperMark evaluates bit-recovery on trigger-conditioned generation (per-sample metric). MiO evaluates population-level ownership verification via paired-delta protocol. Different evaluation primitives; both remain robust under LoRA fine-tuning.

Method	Metric	r= 64	r= 256	Threat Model
SleeperMark (trigger-based)	Bit acc. (WM)	0.9861 ± 0.0275	0.9858 ± 0.0338	γ-recipe LoRA (domain adapt, MSE)
	noWM (control)	0.5071 ± 0.0749	0.5021 ± 0.0728	
MiO (Ours) (population-level)	AUC (membership)	0.995	0.999	γ-recipe LoRA (paired-delta)
	Ownership verdict	✓(ratio $7.46\times$)	✗(ratio $3.62\times$)	

SleeperMark: N=200 SD-v1.4 images per variant, LoRA adapters from adversary fine-tuning on disjoint data.
MiO: N=1000 COCO private images, same threat model (γ-recipe).

Table 9: Aggregation ablation on CIFAR-10. Lower percentiles sharpen separation up to a point; Q25 offers the best trade-off before outlier sensitivity sets in.

Aggregation	Cohen's d	Ratio
Mean	-18.2	15.1
Q10	-21.5	20.8
Q25 (ours)	$\mathbf{-23.93}$	$\mathbf{24.40}$
Median	-19.8	17.2

criteria. The strong separation persists across all three adapter configurations, establishing that the ownership signal remains robust under realistic post-theft fine-tuning scenarios.

5.6 RQ3: Score Stability and Threshold Calibration

Several design choices underlie the verification pipeline—how to collapse a sequence of per-timestep errors into a single score, how many timesteps are needed for that score to be reliable, and whether the Gaussian assumption in our quantile regression is worth the modelling commitment. We address each in turn on CIFAR-10; ensemble size and wall-clock breakdowns appear in Appendix B.

Table 9 compares four ways of summarising the t-error sequence. The mean, which weights every timestep equally, washes out the signal at noise levels where the memorisation gap is widest; the median is slightly more selective but still averages over a broad middle band. Moving to lower percentiles helps because the ownership signal concentrates at moderate noise levels—timesteps at which the model reconstructs its own training data with noticeably lower error than unseen samples. Q25 captures this sweet spot, reaching $|d| = 23.93$ and a $24.4\times$ ratio. Pushing further to Q10 begins to pick up outlier timesteps whose high variance hurts more than their selectivity helps, and separation drops back slightly.

The number of sampled timesteps K matters less than one might expect. Varying K from 10 to 100, we find that $|d|$ plateaus once $K \geq 25$ and changes negligibly beyond $K = 50$. We therefore fix $K = 50$ throughout; practitioners under tighter compute budgets can halve this to $K = 25$ with only marginal loss.

Finally, we ask whether the Gaussian distributional assumption in our quantile regression buys anything over a distribution-free alternative. Pinball-loss QR optimises directly for a target quantile τ without assuming a parametric form, but it requires a separate model for every τ of interest. On CIFAR-10, at $\tau \in \{0.999,\ 0.9999\}$ (FPR $= 0.1\%$ and 0.01%), the two approaches yield comparable TPR. The practical difference is flexibility: Gaussian QR trains once and evaluates any τ in closed form, cutting both the parameter count and the retraining burden roughly in half. For a verifier who may need to assess ownership claims at several significance levels—exactly the setting envisioned by our protocol—this is a meaningful convenience.

6 Discussion and Limitations

We frame scope conditions rather than limitations.

White-box Access: Our method applies where model weights are available, standard in IP litigation (courts compel disclosure), regulatory audits (GDPR/AI Act), partnership disputes, and increasingly common open-weight models (LLaMA, Mistral).

Watermark Retention: Verification requires the owner to retain the watermark set, analogous to keeping receipts for warranty claims.

Baseline Availability: Results are strongest with domain-matched public baselines; for novel domains, general-purpose baselines provide conservative bounds.

Gaussian Assumption: The Gaussian model provides a principled approximation; empirical calibration shows quantile estimates remain accurate under moderate misspecification.

Distillation Boundary: Our threat model assumes the adversary retains the original training distribution via weight-preserving operations (fine-tuning, pruning, quantisation). Knowledge distillation—generating synthetic data from the stolen model and training a fresh student—violates this assumption and verification correctly reports no ownership ($d = -2.93$, FAIL; Appendix G). This is a principled boundary shared by proactive watermarking methods: WDM and Zhao embed signals into model weights, which distillation with fresh initialisation erases equally.

What This Work Does NOT Address: API-only verification (we focus on white-box), full retraining adversaries (our threat model assumes fine-tuning), and adversaries specifically optimizing to remove memorization signals.

7 Conclusion

We presented a novel ownership verification framework for diffusion models using t-error based reconstruction disparity and Gaussian Quantile Regression. By formalizing the distinction between membership inference and ownership verification, we established that public baselines and statistical hypothesis testing are *necessary* components for ownership claims. Our two-point verification criteria (consistency, separation) provide rigorous, reproducible verification. Experiments across four datasets demonstrate Cohen's $d > 18$ separation across all reference models, with criteria satisfied even after adversarial MMD fine-tuning.

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A Additional Experimental Details

A.1 Hyperparameters

Table 10: Training hyperparameters for DDIM models.

Parameter	Value
Total iterations	400,000
Batch size	128
Learning rate	2×10^{-4}
Optimizer	AdamW
Weight decay	0.0
EMA decay	0.9999
Noise schedule	Cosine
Timesteps T	1000
U-Net Architecture	
Base channels	128
Channel multipliers	[1, 2, 2, 2]
Attention resolutions	[16]
Dropout	0.1

Table 11: Gaussian QR training hyperparameters.

Parameter	Value
Backbone	ResNet-18 (pretrained)
Ensemble size B	50
Bootstrap ratio	0.8
Epochs per model	50
Batch size	256
Learning rate	0.001
LR schedule	Cosine annealing
Early stopping patience	10 epochs

A.2 MMD Fine-tuning Configuration

Table 12: MMD fine-tuning hyperparameters.

Parameter	Value
Iterations	500
Learning rate	5×10^{-6}
Batch size	128
DDIM steps	10
t_{start}	900
EMA decay	0.9999
Kernel	Cubic polynomial
Feature extractor	CLIP ViT-B/32 (frozen)

B Additional Ablations

B.1 Ensemble Size

We vary the ensemble size B from 1 to 100 and measure the standard deviation of the predicted quantile $\hat{q}_\tau(x)$ across five independent runs on the CIFAR-10 watermark set at $\tau = 0.999$ (FPR $= 0.1\%$). Table 13 reports the results. Prediction variance drops sharply from $B = 1$ to $B = 20$, continues to decrease up to $B = 50$, and plateaus beyond that point. We adopt $B = 50$ as the default, which offers a favorable trade-off between stability and computational cost. Since ensemble members are trained independently, scaling from $B = 50$ to any larger value is embarrassingly parallel and requires no inter-model communication.

Table 13: Effect of ensemble size B on quantile prediction stability (CIFAR-10, $\tau = 0.999$). Std. dev. computed over 5 independent runs.

Ensemble B	Std($\hat{q}_\tau$)	TPR @ FPR 0.1%
1	0.142	0.83 ± 0.05
10	0.068	0.87 ± 0.03
20	0.041	0.88 ± 0.03
50 (ours)	0.019	0.89 ± 0.03
100	0.016	0.89 ± 0.02

B.2 Computational Cost

Table 14 summarizes the computational requirements of each component. The dominant cost is ensemble training (approximately 100 A100-hours for $B = 50$), which parallelizes trivially. At inference, t-error computation for 5,000 watermark samples takes roughly 10 minutes on a single A100, and ensemble quantile prediction adds less than 1 minute. In total, a single ownership query (from raw images to verdict) completes in under 15 minutes, making the protocol practical for on-demand verification.

By comparison, explicit watermarking methods such as WDM [Peng et al., 2023] require re-training the diffusion model with the watermark objective, which typically takes as long as the original model training (on the order of days for the architectures we consider). MiO's Gaussian QR training is a one-time offline cost, and the owner model itself is never modified.

Table 14: Computational cost breakdown on CIFAR-10 (single NVIDIA A100).

Component	Time	Parameters
DDIM training (owner model)	$\sim$48 h	35.7M
Gaussian QR ($\times$1 model)	$\sim$2 h	11.2M
Gaussian QR ensemble (B=50)	$\sim$100 h†	560M total
Inference (per verification query)		
t-error scoring (5k samples)	$\sim$10 min	—
Ensemble prediction	$<$1 min	—
Two-point criteria	$<$1 s	—
Total query time	**$\sim$11 min**	—

† Embarrassingly parallel; wall-clock $\sim$2 h with 50 GPUs.

C Perceptual Hash Evaluation

We use threshold of 5 bits as a balance between FPR and FNR. Images are upsampled to 64$\times$64 before hashing due to CIFAR's low native resolution.

D Threshold Sensitivity Analysis

All datasets achieve $d > 18$, providing substantial margin above our recommended threshold.

Table 15: Perceptual hash (pHash) performance on CIFAR-10 train vs. test set deduplication. FPR: false positive rate (distinct images flagged as duplicates). FNR: false negative rate (actual duplicates missed).

Hamming Threshold	FPR (%)	FNR (%)	Flagged Pairs
3 bits	0.02	5.1	12
5 bits (ours)	0.10	2.0	48
7 bits	0.35	0.8	175

Table 16: Verification pass rate across threshold choices. All datasets pass even with stricter thresholds.

Threshold	CIFAR-10	CIFAR-100	STL-10	CelebA
$d > 1.0$	✓	✓	✓	✓
$d > 1.5$	✓	✓	✓	✓
$d > 2.0$ (ours)	✓	✓	✓	✓
$d > 2.5$	✓	✓	✓	✓
$d > 3.0$	✓	✓	✓	✓

E Gaussian QR Training Details

Algorithm 3 Gaussian Quantile Regression Training

Require: Auxiliary non-member data $\mathcal{D}_{\text{aux}}$, ensemble size B, bootstrap ratio r
Ensure: Trained ensemble $\{f_\theta^{(b)}\}_{b=1}^B$
  for $b = 1$ to B **do**
    $\mathcal{D}_b \leftarrow$ BootstrapSample$(\mathcal{D}_{\text{aux}}, r)$ {80% with replacement}
    $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}} \leftarrow$ Split$(\mathcal{D}_b, 0.9)$
    Initialize $f_\theta^{(b)}$ with ResNet-18 backbone
    repeat
      for batch $(x, s(x), \phi(x))$ in $\mathcal{D}_{\text{train}}$ **do**
        $y \leftarrow \log(1 + s(x))$
        $(\mu, \log\sigma) \leftarrow f_\theta^{(b)}(x, \phi(x))$
        $\mathcal{L} \leftarrow (y - \mu)^2/(2\sigma^2) + \log\sigma$
        Update θ via gradient descent
      end for
    until early stopping (patience = 10)
  end for
  return $\{f_\theta^{(b)}\}_{b=1}^B$

F Retroactive Claim Defense

An adversary who observes a published model could attempt to retroactively fabricate an ownership claim by selecting a subset of training data that happens to exhibit low t-error. Table 17 evaluates this attack under several adversary strategies on CIFAR-10.

G Threat Model Boundary: Distillation

An adversary who generates a large synthetic dataset from the stolen model and trains a new model from scratch effectively performs *knowledge distillation*. Because the student is initialised independently and never sees the owner's training data directly, per-example memorisation of $\mathcal{W}$ is not expected to transfer.

Table 17: Retroactive claim defense on CIFAR-10. An adversary attempts to construct a fake watermark set $\mathcal{W}'$ that passes the two-point verification protocol. Cohen's d is computed against public baselines. Ratio column is reported for reference only and is no longer a verification criterion. Two-point: both verification criteria (Sec. 4.4) satisfied (✓) or not (✗).

Adversary Strategy	Cohen's d	Ratio (ref.)	Two-point
True owner ($\mathcal{W}$, 5K)	24.07	24.5×	✓
Random subset (avg. of 100)	16.97 ± 0.17	6.2×	hash fail
Cherry-picked top-5K	25.49	50.8×	hash fail
Sophisticated adversary	25.49	50.8×	hash fail
Non-member (test set, 5K)	15.18	6.3×	✗
Wrong model	0.00	1.0×	✗

Experiment. We generate 50,000 synthetic CIFAR-10 images from Model A via 10-step DDIM sampling and train a student model from scratch (100,000 iterations, batch size 128, fresh random initialisation). Table 18 reports t-error statistics on $\mathcal{W}$.

Table 18: Distillation boundary experiment on CIFAR-10. The student model, trained on synthetic data from the owner, exhibits $3.72\times$ higher t-error than the owner and fails the consistency criterion.

Model	Mean t-error	Std	Ratio vs. Owner
Model A (owner)	28.71	10.85	1.0×
Student (distilled)	106.90	36.14	3.72×
ddpm-cifar10 (reference)	703.84	38.50	24.5×
Random UNet (lower bound)	3037.31	65.75	105.8×

The student's mean t-error on $\mathcal{W}$ is 106.9, compared to 28.7 for the owner—a $3.72\times$ gap (Cohen's $d = -2.93$, $p \approx 0$). The verification protocol correctly returns FAIL: the consistency criterion rejects because the student does not reproduce the owner's per-example memorisation pattern.

This outcome is expected and represents a principled boundary of the method rather than a failure mode. Distillation with fresh initialisation approximates full retraining: generating a sufficiently large synthetic dataset from a diffusion model and training a new model from scratch approaches the computational cost of the original training procedure, undermining the economic motivation for model theft. Critically, the protocol does not produce a false positive—it correctly reports that ownership cannot be established, which is the far more important guarantee.

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